

- This exam contains 8 pages (including this cover page) and 13 problems. Check to see if any pages are missing.
- This is the fourth exam and will count for 15% of your final grade. You have 90 minutes to complete this exam.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work when work is required will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 15 bonus points.
- Do not write in the $\checkmark$ + - box, because this is for grading. The grading scale is:
 $\checkmark = 100\%$, $+$ = 50%, $-$ = 0%

Page:	3	4	5	6	7	Total
Points:	20	10	20	20	30	100
Score:						

1. (10 points) Woot! You won the PowerBall lottery of 189 million dollars! (payable in 30 equal installments over 29 years) Since less than the \$189M is available, the PowerBall officials deposit some value into an account that pays 4% APY; you get one payment up front, then one payment per year thereafter for 29 years. How much needs to be in the account to make the appropriate payments?

(HINT: Even though you've gained money yourself, this question is about PowerBall *paying off the balance of your winnings.*)

2. (10 points) If you borrow some money from a payday loan company that charges 350% APR compounded daily and pay back \$1457.43 in total (in regular daily payments) to close out the loan 13 days later, how much did you borrow?

3. (10 points) If you put \$250 per month into an account paying 4% APR compounded monthly, how long will it be until you have \$1,500,000 in the account?

✓ + −

HINT: Recall that in section 10.3, the future value of \$ P earning a periodic interest at a rate of $p\%$ compounded periodically for T periods was given by the formula $F = P(1 + p)^T$. Now, note that the amortization formula

$$M = P \frac{p(1 + p)^T}{(1 + p)^T - 1}$$

Can be re-written as

$$M = F \frac{p}{(1 + p)^T - 1}.$$

Solving for F and T respectively, this also yields

$$F = M \frac{(1 + p)^T - 1}{p}, \quad T = \frac{\ln \left(\frac{Fp}{M} + 1 \right)}{\ln (p + 1)}.$$

4. (10 points) Kendall has run up his credit card to \$8,000. He pays 18% APR compounded monthly. How much does he need to pay per month to eliminate the balance after 2 years?

5. (10 points) If you pay \$250 per month towards a \$20,000 loan you took at 4% APR compounded monthly, how long will it be until you pay off the balance?

6. (10 points) Starting at the age of 20, at the end of each month, you invest \$250 into a retirement IRA (individual retirement account). The IRA guarantees an APR of 3% compounded monthly for the life of the account. How much money will you have in the account when you turn 55? Give your answer as a decimal rounded to the nearest hundredth.

7. (10 points) Suppose you run up a balance of \$15,000 on a credit card, with an interest rate of 17% compounded monthly. If you only pay \$150.50 per month, how much interest will you pay in total by the time you pay off the balance?

8. (10 points) Give the formula for the N th Fibonacci number in terms of previous Fibonacci numbers and write the first eight numbers in the Fibonacci sequence.

☐ ✓ ☐ + ☐ −

9. (10 points) State the golden property and write the exact value of the golden ratio (not a decimal approximation of it).

☐ ✓ ☐ + ☐ −

10. (10 points) Write the exact value of ϕ^2 (not a decimal approximation of it).

☐ ✓ ☐ + ☐ −

11. (5 points (bonus)) Write the value of ϕ^{20} in terms of Fibonacci numbers.

☐ ✓ ☐ + ☐ −

12. (5 points (bonus)) Write the most accurate approximation of the golden ratio possible using only division and the first 100 Fibonacci numbers.

13. (5 points (bonus)) You are considering two savings accounts to put \$600 in to per month for 12 months. Account A pays 9.72% APR compounded monthly. Account B pays $x\%$ APR compounded semi-annually. Solve for x so that the same amount of money would be in either account at the end of 10 months.

(HINT: "Semi-annually" means "every 6 months")